

A μeV string-axion target with an f_a -invariant effective coupling, inside the FLASH band

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Abstract

We present one concrete, falsifiable coordinate for the axion dark-matter program, derived within an independent braneworld cosmology (Oscillating Brane Theory, OBT) whose type-IIB LVS completion fixes the string scale $M_s = 1.19 \times 10^{12}$ GeV and contains an ultra-light Kähler-axion mode at the μeV scale ($\nu \simeq 242$ MHz central; order-of-magnitude bracket $\sim 0.3\text{--}3$ μeV). The photon coupling is $g_{a\gamma\gamma} = (\alpha/2\pi) C/f_a$ with $f_a \sim M_s$ natural, and the standard misalignment relic gives $\Omega_a h^2 \simeq 1.2 \times 10^{-3} (f_a/M_s)^2 \theta_i^2$ — 1% of the dark matter at the natural point, 100% at $f_a \simeq 1.2 \times 10^{13}$ GeV. Our main result is that this decay-constant uncertainty does not propagate to the experiment: because the string-ALP mass is instanton-set (independent of f_a , unlike the QCD axion), the density-weighted haloscope coupling is exactly f_a -invariant below saturation, $g_{\text{eff}} \equiv g\sqrt{\Omega_a/\Omega_{\text{DM}}} = (\alpha/2\pi) C\theta_i/f_{a,\text{DM}} = 9.8 \times 10^{-17} C\theta_i \text{ GeV}^{-1}$. The flat floor crosses the QCD band inside the target decade and sits five orders of magnitude below the CAST bound. The proposed FLASH haloscope (100–300 MHz) contains the central target: its declared phase-1 sensitivity grazes the floor ($0.98\times$), and its phase-2 goal cuts a factor ~ 5 below it — decisive for the entire bracket at $C, \theta_i = \mathcal{O}(1)$. We state the falsification condition in advance: excluding the μeV decade at $g_{\text{eff}} \lesssim 1 \times 10^{-17} \text{ GeV}^{-1}$ kills the whole misalignment chain at $\mathcal{O}(1)$ angles.

1 Status and provenance (read first)

This note comes from *Oscillating Brane Theory* (OBT), an independent-researcher braneworld cosmology (the universe as a 4D membrane oscillating in a 5D AdS bulk; not peer-reviewed; the full document and validation suite are public at the repository above). The axion chain below belongs to its *quarantined exploration layer* (internally labeled V9.0): it is not claimed with the confidence of the framework’s core phenomenology, and the epistemic posture of this note is deliberately modest. **We present one concrete, falsifiable coordinate in open parameter space, with its complete derivation chain and every assumption listed.** The community can price the theory however it likes; the coordinate is cheap to test, and the falsification statement is clean either way.

The target in one line: a string ALP at the μeV scale ($\nu \simeq 242$ MHz central; order-of-magnitude bracket $\sim 0.3\text{--}3$ μeV) whose density-weighted haloscope coupling is pinned, *independently of its decay constant*, at $g_{\text{eff}} \simeq 1.0 \times 10^{-16} C\theta_i \text{ GeV}^{-1}$ — grazing the declared FLASH phase-1 sensitivity and a factor ~ 5 above its phase-2 goal.

*Independent researcher; provencal.romain@teleadmin.net. All supporting calculations are runnable scripts in the public repository, <https://github.com/Teleadmin-ai/oscillating-brane-DM> (directory `explorations/qubit_sensor/`); theory documents at <https://higgs-cosmology.com/>.

2 The derivation chain

Each step corresponds to a runnable script in the repository (Sec. 8).

1. **The scale.** OBT’s UV completion is a Klebanov–Strassler / LVS type-IIB compactification whose flux integers ($K = 21$, $M = 10$, $g_s = 0.1$) reproduce, top-down and with no fine-tuning, the brane tension $\tau_0^{1/3} = 257 \text{ MeV} \approx \Lambda_{\text{QCD}}$ that the cosmological fit independently requires (a bottom-up/top-down convergence). The resulting LVS spectrum contains the string scale $M_s = M_{\text{Pl}}/\sqrt{\mathcal{V}} = 1.19 \times 10^{12} \text{ GeV}$ (volume $\mathcal{V} \approx 4 \times 10^{12}$) and an ultra-light modulus at $m_V \sim 10^{-6} \text{ eV}$ (order of magnitude). This μeV scale predates the axion chain; it was not tuned toward any experiment.
2. **The identification** (`mv_coupling.py`). The μeV mode is identified as the imaginary (axion) partner of a blow-up/fibre Kähler modulus: the shift symmetry makes it ultra-light with *derivative couplings only* (no static fifth force, evading Eötvös/Cassini bounds), and the photon coupling is

$$g_{a\gamma\gamma} = \frac{\alpha}{2\pi} \frac{C}{f_a}, \quad C = \mathcal{O}(1) \text{ (unknown anomaly coefficient)}. \quad (1)$$

3. **The decay constant is not free** (`mv_fa_lvs.py`). The LVS axiverse [2] brackets $f_a \in [\sim 10^{10}, \sim 10^{13}] \text{ GeV}$; a blow-up/fibre axion sits naturally at $f_a \sim M_s \approx 1.2 \times 10^{12} \text{ GeV}$, giving $g_{a\gamma\gamma} \approx 9.8 \times 10^{-16} C \text{ GeV}^{-1}$.
4. **The relic abundance** (`mv_abundance.py`; standard misalignment [3]). The field oscillates early ($T_{\text{osc}} \approx 16 \text{ GeV}$, above the QCD scale), giving

$$\Omega_a h^2 \simeq 1.2 \times 10^{-3} \left(\frac{f_a}{M_s} \right)^2 \theta_i^2, \quad (2)$$

i.e. $\sim 1\%$ of the dark matter at the natural point, rising to 100% at $f_a \simeq 1.2 \times 10^{13} \text{ GeV}$ (the LVS high end; over-closure forbids larger f_a at $\theta_i \sim 1$). In OBT the *dominant* dark matter is geometric (a projected-Weyl braneworld effect), so a sub-dominant axion is internally consistent, and the $\sim 1\%$ fraction suppresses axion-CDM isocurvature by $\sim 10^{-4}$ — safe for a wide range of inflation scales (the full-DM corner instead pushes toward low-scale inflation).

3 Main result: the f_a -invariant effective coupling

A haloscope measures the power $g^2 \rho_a$, i.e. the effective coupling $g_{\text{eff}} = g \sqrt{\Omega_a / \Omega_{\text{DM}}}$ (assuming the axion fraction clusters like the bulk dark matter). In this chain $g \propto 1/f_a$ while $\Omega_a \propto f_a^2$ — because the string-ALP mass is instanton-set, *independent of f_a* (unlike the QCD axion) — so the f_a dependence cancels exactly below saturation:

$$g_{\text{eff}} = \frac{\alpha}{2\pi} \frac{C \theta_i}{f_{a,\text{DM}}} = 9.8 \times 10^{-17} C \theta_i \text{ GeV}^{-1}, \quad f_{a,\text{DM}} = 1.2 \times 10^{13} \text{ GeV}, \quad (3)$$

verified numerically to 10^{-9} across the three-decade LVS bracket (`mv_effective_coupling.py`). **The apparent f_a uncertainty (a 1%–100% dark-matter fraction) does not propagate to the experiment: every point of the bracket presents the same haloscope target.** The residual theory spread is the honest $\mathcal{O}(1)$ pair $C\theta_i$ and the order of magnitude on the μeV mass itself.

4 Where the target sits

m_a (μeV)	ν (MHz)	g_{KSVZ} (GeV^{-1})	g_{DFSZ} (GeV^{-1})	OBT g_{eff} ($C\theta_i=1$)
0.30	73	1.2×10^{-16}	4.6×10^{-17}	9.8×10^{-17}
0.64	155	2.5×10^{-16}	9.8×10^{-17}	9.8×10^{-17}
1.00	242	3.9×10^{-16}	1.5×10^{-16}	9.8×10^{-17}
1.49	360	5.8×10^{-16}	2.3×10^{-16}	9.8×10^{-17}
3.00	725	1.2×10^{-15}	4.6×10^{-16}	9.8×10^{-17}

Table 1: The flat OBT floor vs. the QCD band ($m_a = 5.70 \mu\text{eV} \times 10^{12} \text{GeV}/f_a$ [5]; KSVZ $|C_\gamma| = 1.92$, DFSZ $|C_\gamma| = 0.75$).

- The flat floor **crosses the QCD band inside the target decade** (above DFSZ below $0.64 \mu\text{eV}$, above KSVZ below $0.25 \mu\text{eV}$). It is a *string ALP near, not on, the QCD relation*: the QCD-axion mass at $f_a = M_s$ would be $4.8 \mu\text{eV}$, within $\sim 5\times$ of m_V .
- **Helioscopes/astrophysics**: CAST ($6.6 \times 10^{-11} \text{GeV}^{-1}$ [10]) constrains g , not g_{eff} ; the *entire* bracket sits ≥ 2.8 orders below it (the natural corner 4.8 orders). The window is genuinely open.
- **Quantum-hardware aside**: we computed (`axion_photonic_chip.py`) that photonic-chip-class quantum processors *cannot* see this field (16–18 orders short; the optical-cavity storage-time cap gains only $\sim 60\times$); μeV axions are resonant microwave-cavity + magnet territory — this note’s target is for haloscopes.

5 The experimental match: FLASH and the μeV -decade program

The proposed **FLASH** haloscope (FINUDA magnet, Frascati [6]) is declared for 100–300 MHz (QCD-axion masses 0.49 – $1.49 \mu\text{eV}$) — the OBT central target (242 MHz) sits *inside* that band:

- **FLASH phase 1** (microstrip SQUID, DFSZ-class $\sim 1 \times 10^{-16} \text{GeV}^{-1}$): the OBT floor is $0.98\times$ that depth — phase 1 *grazes the whole target*.
- **FLASH phase 2** (100 mK, $\sim 2 \times 10^{-17} \text{GeV}^{-1}$): cuts a factor ~ 5 below the floor — decisive for the entire f_a bracket at $C\theta_i = \mathcal{O}(1)$.
- The wings of the mass bracket (0.3 – 0.5 and 1.5 – $3 \mu\text{eV}$, i.e. 73–120 and 360–725 MHz) fall to the neighboring programs (lumped-element / DM-Radio-class below; the historic and next-generation cavity program above [7, 8], where reaching the floor means DFSZ/2 to DFSZ/5 depth). Qubit single-photon counting [9] is the readout class that accelerates such scans.

6 Falsifiability (declared before any data)

- **Exclusion**: a scan of the ~ 0.3 – $3 \mu\text{eV}$ decade at $g_{\text{eff}} \lesssim 1 \times 10^{-17} \text{GeV}^{-1}$ ($\approx \text{floor}/10$) **kills the entire misalignment chain at $\mathcal{O}(1)$ angles** — the OBT axion bone dies cleanly

(the theory’s core is separate; this falsifies its one derived direct-detection channel). FLASH phase 2 alone already bites the central half-decade at floor/5.

- **Detection:** a signal at μeV masses with g_{eff} within $\sim\times 3$ of $10^{-16} \text{ GeV}^{-1}$ matches the chain; the OBT-vs-QCD-axion discriminant is then the (m, g) position relative to the QCD relation (the flat floor is mass-independent; the QCD band scales $g \propto m$ — distinguishable away from the 0.25–0.64 μeV crossings). A detection *on* the QCD relation would remain ambiguous — we state this plainly.
- **What we will not do:** claim support from a null that only grazes the floor, or dress an anomaly elsewhere as this field. (Our own quantum-hardware protocol pre-registers the same discipline.)

7 Open assumptions (all stated)

1. **Axion-vs-saxion identification** of the μeV LVS mode (the chain’s weakest link; the saxion partner is heavier, but the identification is an assumption, not a derivation).
2. **The μeV mass itself is order-of-magnitude** (an LVS spectrum scale, not a sharp prediction) — hence the decade-wide mass bracket.
3. C and θ_i are $\mathcal{O}(1)$ unknowns (the floor scales linearly in each).
4. **Invariance assumptions:** m_a independent of f_a (string ALP); standard misalignment with no post- T_{osc} entropy injection; below over-closure saturation; local axion fraction proportional to the bulk fraction.
5. $f_a \sim M_s$ natural within a factor $\sim \mathcal{O}(10)$ (2π ’s, volume suppressions) — *absorbed* by the invariance for g_{eff} , but relevant to the raw g if a density-independent probe is used.

8 Supporting calculations

All in `explorations/qubit_sensor/` of the repository, runnable as-is: `mv_coupling.py` (identification + g), `mv_abundance.py` (T_{osc} , $\Omega_a h^2$, the isocurvature trade-off), `mv_fa_lvs.py` (the LVS f_a bracket), `mv_effective_coupling.py` (the f_a -invariance, QCD-band crossings, FLASH/CAST margins), `chi_alp_sensitivity.py` (why the quantum-sensor mass window selects this mode), `axion_photonic_chip.py` (why photonic chips cannot see it). Core-framework scales (M_s , the LVS spectrum): `theory.md` on the repository.

Scope note: this proposal detects (or excludes) a light bulk-sector field of the theory. It is one of OBT’s three falsifiable near-term “bones” (with the evolving MOND scale $a_0(z) = cH(z)/2\pi$ for Euclid/Rubin, and a 5D-enhanced Penrose–Diósi collapse size-scan for levitated optomechanics).

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References

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